Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
12.	(a)			$\frac{1.44}{18.02}$ = 0.080 mol (1) e = $\frac{0.080}{0.020}$ = 4 (1)	2		1	1	1	
	(b)			$\frac{\frac{490}{1000}}{24.5}$ = 0.020 (1) c = $\frac{0.020}{0.020}$ = 1 (1)	2		1	1	1	
	(c)			white part of precipitate caused by $Mg(OH)_2$ produced from Mg^{2+} ions (1) grey-green precipitate so X is chromium/Cr (1) precipitate dissolves because Cr^{3+} is amphoteric (1)	3	1 1		1		1 1 1
	(d)			moles of HCl added = $\frac{25}{1000}$ × 1.00 = 0.025 mol (1) moles of NaOH required to neutralise remaining acid = $\frac{14}{1000}$ × 0.500 = 0.007 mol moles of HCl remaining = 0.007 mol (1) moles of HCl reacting with $OH^–$ and $CO_3^{2–}$ = 0.025 – 0.007 = 0.018 mol (1) Moles of HCl reacting with $OH^–$ = 0.018–0.002 = 0.016 mol d = $\frac{0.016}{0.001}$ = 16 (1) ecf possible from part (b) for final mark	4		1	1 1 1	1 1 1	1 1